



CHAPTER-6



HALOALKANES AND HALOARENES

★ 1. INTRODUCTION

- Haloalkanes are **alkyl** derivatives in which one or more H-atom of alkane is replaced by **halogen** (F, Cl, Br, I).
- Haloarenes are **aryl** derivatives in which one or more H-atom of **arene (benzene)** is replaced by halogen.

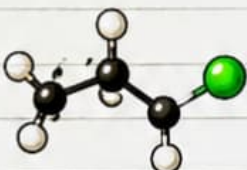
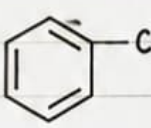
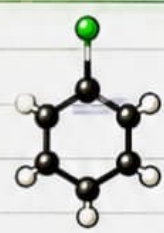
General Formula

Haloalkane $\rightarrow R-X$

Haloarene $\rightarrow Ar-X$

(X = F, Cl, Br, I)

Examples

Haloalkanes	Haloarenes
CH_3-CH_2-Cl (Chloroethane)  (3D Structure)	 (Chlorobenzene)  (3D Structure)

Key Points



- C-X bond is polar.
- Reactivity depends on nature of halogen.
- Order: $R-F < R-Cl < R-Br < R-I$

2. NOMENCLATURE

Haloalkanes

Name of alkyl group + prefix (**fluoro-**, **chloro-**, **bromo-**, **iodo-**) + alkane

Example:

- $CH_3-CH_2-Br \rightarrow$ Bromoethane
- $CH_3-CHCl-CH_3 \rightarrow$ 2-Chloropropane

Haloarenes

Name of halogen + benzene

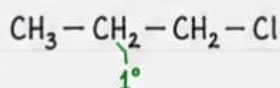
Example:

- $C_6H_5-Cl \rightarrow$ Chlorobenzene

3. CLASSIFICATION OF HALOALKANES

Primary (1°)

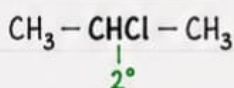
Halogen attached to 1° carbon



(1-Chloropropane)

Secondary (2°)

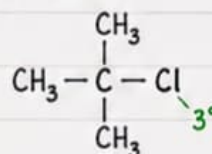
Halogen attached to 2° carbon



(2-Chloropropane)

Tertiary (3°)

Halogen attached to 3° carbon



(2-Chloro-2-methylpropane)

Stability Order

$3^\circ > 2^\circ > 1^\circ$

Reason: +I effect of alkyl groups stabilizes the carbocation intermediate.

4. PHYSICAL PROPERTIES

- Higher members are colourless, insoluble in water but soluble in organic solvents.
- Boiling point** is **higher** than corresponding alkanes but lower than alcohols.
- B.P. increases** with **increase in molecular mass**.

Boiling Point Comparison ($^\circ C$)

CH_3-CH_3 (Ethane)	CH_3-CH_2-Cl (Chloroethane)	CH_3-CH_2-OH (Ethanol)
-89	< 12	< 78

Dipole Moment

C-X bond is polar $\rightarrow$ haloalkanes have greater dipole moment than alkanes.



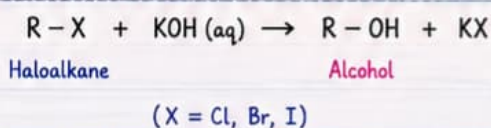
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5. CHEMICAL PROPERTIES OF HALOALKANES

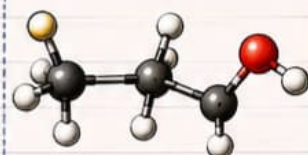
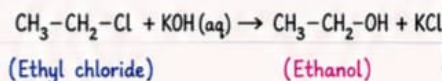
1. Nucleophilic Substitution Reactions

In these reactions, a **nucleophile** replaces the **halogen** atom in haloalkanes.

(a) With aqueous KOH (or NaOH) → Formation of Alcohols

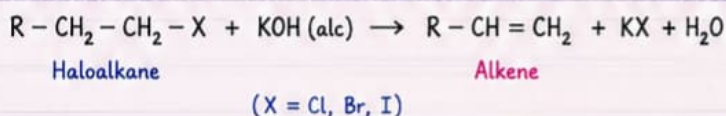


Example:

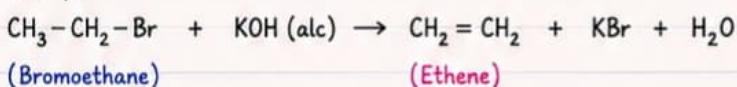


Ethanol (3D Structure)

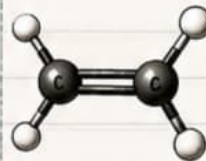
(b) With alcoholic KOH → Formation of Alkenes (Elimination)



Example:



Ethene (3D Structure)

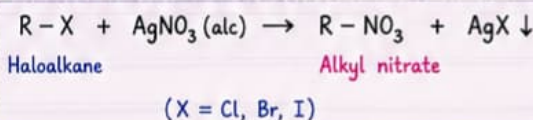


Key Point

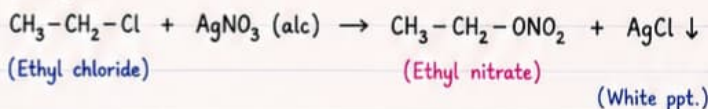
Alcoholic KOH →
Elimination
(Alkene)

Aqueous KOH →
Substitution
(Alcohol)

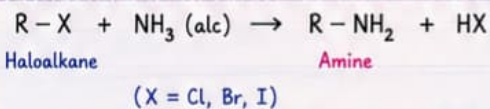
(c) With silver nitrate in ethanol → Formation of haloalkanes (precipitation)



Example:



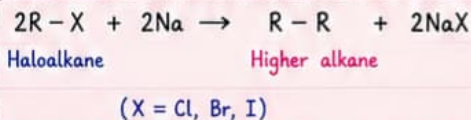
2. Nucleophilic Substitution in Presence of Ammonia



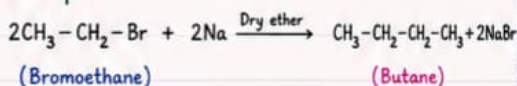
Example:



3. Wurtz Reaction (Coupling Reaction)



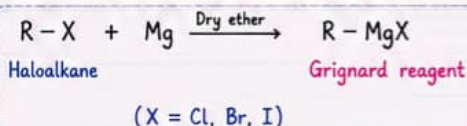
Example:



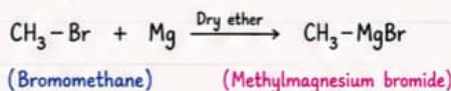
Conditions

- Sodium metal
- Dry ether as solvent
- Alkyl halide must be primary (1°) for best results.

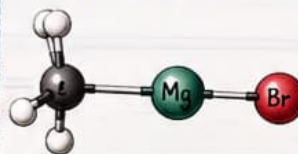
4. Formation of Grignard Reagents



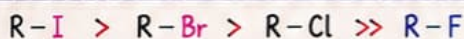
Example:



Grignard reagent (3D View)



Reactivity of Haloalkanes (General Order)



Reason:

C-X bond strength: $\text{R-I} < \text{R-Br} < \text{R-Cl} \ll \text{R-F}$
(weaker bond → higher reactivity)

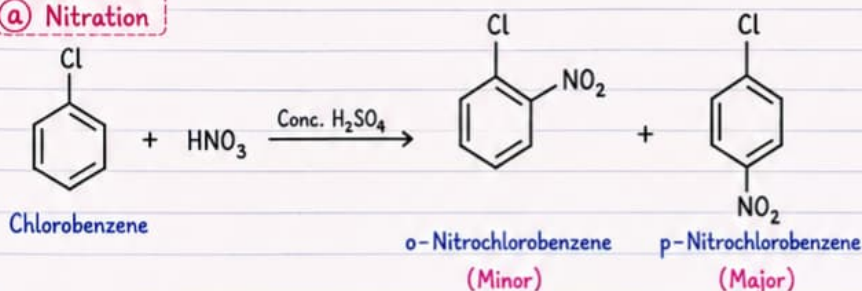
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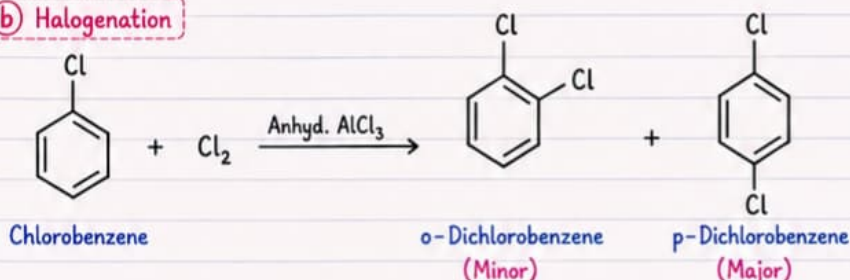
6. REACTIONS OF HALOARENES (ELECTROPHILIC SUBSTITUTION)

- In haloarenes, halogen is **deactivating** but **o-, p-** directing.

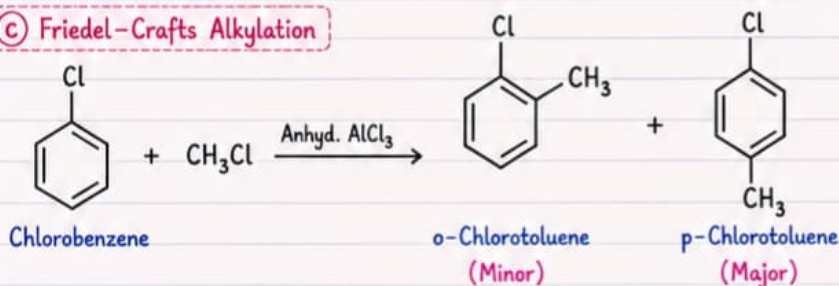
(a) Nitration



(b) Halogenation



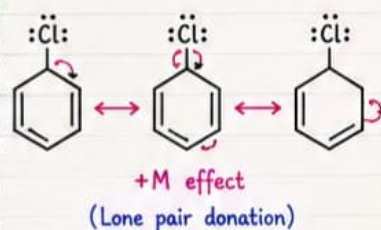
(c) Friedel-Crafts Alkylation



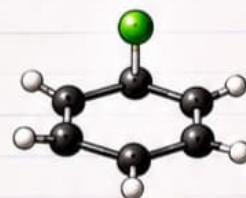
Key Points

- C-X bond in haloarenes has partial double bond character (resonance).
- Hence, haloarenes are less reactive than benzene.
- Halogen is **o-, p-** directing due to resonance.

Resonance in Haloarenes



3D Structure of Chlorobenzene



- Chlorine lies in the plane of benzene ring.

7. USES OF HALOALKANES AND HALOARENES

Haloalkanes

- Used as solvents (e.g., CCl_4 , CHCl_3).
- Refrigerants (e.g., CCl_2F_2 - Freon).
- Fire extinguishers (e.g., Halon - CF_2Cl_2).
- Anesthetics (e.g., Chloroform - CHCl_3).
- Starting materials in organic synthesis.



Haloarenes

- Used in manufacture of dyes, drugs and pesticides.
- Chlorobenzene is used as a solvent and in making other chemicals.
- Iodoform (CHI_3) is used as antiseptic.



Remember

- Haloalkanes $\rightarrow$ more reactive (due to polar C-X bond).
- Haloarenes $\rightarrow$ less reactive (due to resonance).
- Reactivity Order (RX): $\text{RI} > \text{RBr} > \text{RCl} \gg \text{RF}$

8. SUMMARY TABLE

Feature	Haloalkanes	Haloarenes
General Formula	R-X	Ar-X
Nature of C-X bond	Polar	Partial double bond character (due to resonance)
Reactivity	More reactive	Less reactive
Type of reactions	Nucleophilic substitution ($\text{S}_{\text{N}}1$, $\text{S}_{\text{N}}2$)	Electrophilic substitution
Direction in EAS	—	o-, p- directing
Examples	$\text{CH}_3\text{-Cl}$, $\text{CH}_3\text{-Br}$, $\text{CH}_3\text{-I}$	$\text{C}_6\text{H}_5\text{-Cl}$, $\text{C}_6\text{H}_5\text{-Br}$, $\text{C}_6\text{H}_5\text{-I}$

Quick Revision

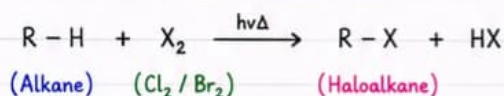
- Haloalkanes $\rightarrow \text{S}_{\text{N}}$ reactions.
- Haloarenes $\rightarrow \text{EAS}$ reactions.
- C-X bond polarity decreases down the group.
- Halogens are o-, p- directors but deactivating in benzene ring.

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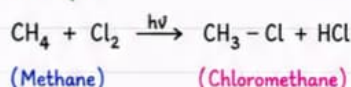
HALOALKANES AND HALOARENES

9. SYNTHESIS (PREPARATION)

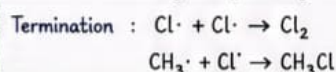
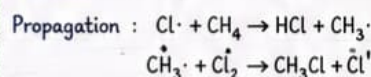
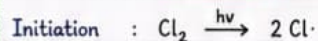
(a) From Alkanes (Free Radical Substitution)



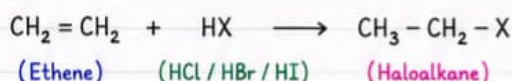
Example:



Free radical mechanism (Chain)



(b) From Alkenes (Electrophilic Addition)



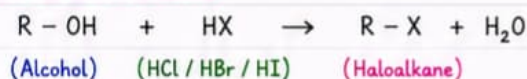
Example:



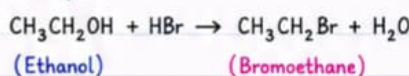
Markovnikov's Rule

In unsymmetrical alkenes, H attaches to the carbon having more H-atoms and X attaches to the carbon having less H-atoms.

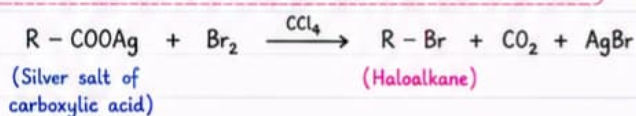
(c) From Alcohols



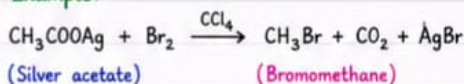
Example:



(d) From Carboxylic Acids (via Hunsdiecker Reaction)



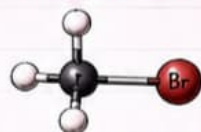
Example:



10. DIFFERENCE BETWEEN HALOALKANES AND HALOARENES

Feature	Haloalkanes (R-X)	Haloarenes (Ar-X)
Bond type	C(sp ³) - X (σ bond)	C(sp ²) - X (partial double bond character due to resonance)
Reactivity	More reactive	Less reactive
Type of reactions	Nucleophilic substitution (S _N 1, S _N 2)	Electrophilic substitution
Influence of halogen	-I effect dominates	+R (resonance) with -I effect
Cleavage	C-X bond breaks easily	C-X bond does not break easily
Example	CH ₃ -Cl, CH ₃ -Br	C ₆ H ₅ -Cl, C ₆ H ₅ -Br

3D Structures



Bromomethane (Haloalkane)



Bromobenzene (Haloarene)

11. IMPORTANT POINTS (NEET FOCUS)

- Reactivity order of haloalkanes : $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$
- Stability of carbocation : $3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$
- Leaving group ability : $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
- Aryl halides are not prepared by nucleophilic substitution because the C-X bond has partial double bond character.
- Haloarenes undergo electrophilic substitution (EAS) such as nitration, halogenation, sulphonation, Friedel-Crafts reactions.

Electrophilic Substitution in Haloarenes (Example)

Nitration of Chlorobenzene



QUICK REVISION

- ✓ Haloalkanes → S_N reactions
- ✓ Haloarenes → EAS reactions
- ✓ Halogens are o-, p- directors
- ✓ C-X bond polarity decreases down the group

Most Important

- ✓ Know the reactivity order.
- ✓ Practice mechanisms.
- ✓ Focus on mechanisms & preparations for NEET.

Memory Tip

"ORCLF" - Reactivity Order
 (Order of Reactivity for haloalkanes)
 $\text{I} > \text{Br} > \text{Cl} > \text{F}$